

6.1.2.

$$(1) R_{p1} = \frac{1}{2} (R_{f1} + R_{f2}) = 1.5 k\Omega$$

(2) ① 当 $R_{p2} = R_p = R_{p3} = 0$ 时.

$$f_0 = \frac{1}{2\pi RC} = 995 \text{ Hz}$$

② 当 $R_{p2} = R_p = R_{p3} = 14.4 k\Omega$ 时

$$f_0 = \frac{1}{2\pi C(R+R_p)} = 99.5 \text{ Hz}$$

$\therefore$ 范围为 $99.5 \sim 995 \text{ Hz}$.

6.1.3.

不可以.

集成运放同相、反相接线应对调

同时放大倍数不足 $A_u = 1 + \frac{R_f}{R_1} \approx 1.84 < 3$.

可以将 $18 k\Omega$ 与 $10 k\Omega$ 互换此时 $A_u' = 3.31 > 3$.

6.1.4.

(a) 可以

(b) 不可以.

发射极左侧 - 侧是 L_1 (感抗), 另一侧是 C (容抗)

(c) 不可以.

发射极左侧 - 侧是 L_1 (感抗), 另一侧是 C_1 (容抗)

(d) 不可以

发射极左侧 - 侧是 L_2 (感抗), 另一侧是 C (容抗)

6.2.1

$$f_0 = \frac{1}{T} = \frac{1}{2RC \ln(1 + \frac{2R_1}{R_2})} = \frac{1}{2 \times 10^3 \times 0.1 \times 10^{-6} \ln(1 + \frac{2}{3})} \approx 7.2 \text{ kHz}$$

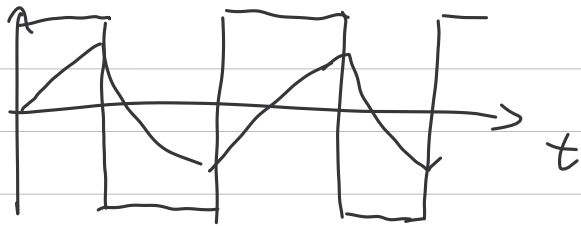
6.2.2

$$f_0 = \frac{1}{2RC \ln(1 + \frac{2R_1}{R_2})} = \frac{1}{2 \times 2 \times 10^3 \times 0.22 \times 10^{-6} \ln(1 + \frac{2 \times 4}{30})} \text{ Hz} = 1.189 \text{ Hz}$$

不考虑 R_2 及 R 电流 $I_{D0} = \frac{U_{om} - U_Z}{R_{D0}} = 1.33 \text{ mA}$

考虑 R_2 及 R 电流 $I_{R2} = \frac{U_Z}{R_2 + R_2} = 0.093 \text{ mA}$

u



$$\bar{I}_{pm} = \frac{U_z - U_{+L}}{R} = \frac{1}{R} \left(U_z - \frac{-U_z R_1}{R_1 + R_2} \right) = 3.61 \text{ mA}$$

$$I_{D0} = I_{D2} + \bar{I}_{pm} + \bar{I}_z = 9.7 \text{ mA}$$

$$R_0 = \frac{U_{om} - U_z}{I_0} = 0.82 \text{ k}\Omega$$

0.2.3

$$T = T_1 + T_2 = RC \ln \left(1 + \frac{2R_1}{R_2} \right) + RC \ln \left(1 + \frac{2R_1}{R_2} \right)$$

$$\eta = \frac{T}{2T} = 50\%$$

$$\begin{aligned} T = T_1 + T_2 &= (R_p' + R) C \ln \left(1 + \frac{2R_1}{R_2} \right) + (R_p' + R) C \ln \left(1 + \frac{2R_1}{R_2} \right) \\ &= (156 + 2 \times 5) \times 10^3 \times 0.1 \times 10^{-6} \times \ln \left(1 + \frac{2 \times 80}{100} \right) \\ &= 6.6 \text{ ms} \end{aligned}$$

$$f_0 = 152 \text{ Hz}$$

① 当 R_p 在最上时. $T_{1.2} = 0.5 \text{ ms}$

当 R_p 在最下时 $T_{1.2} = 6.1 \text{ ms}$

$$\therefore \eta = \frac{0.5}{6.6} \sim \frac{0.1}{6.6} \text{ 即 } 7.6\% \sim 92.4\%$$